The Industrial Stack Behind Artificial Intelligence

Where Economic Rent Survives Across the AI Infrastructure Chain, 2026–2036

Data current to Q2 2026. Figures approximate, from public filings and industry sources, labelled by confidence. Geography is treated as manufacturing concentration and supply-chain risk only.

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1. Executive conclusion

AI is a physical industrial system — models, accelerators, semiconductor manufacturing, memory, packaging, electricity, power conversion, grid, generation, storage. The binding constraint has moved down that chain into power, but constraint location does not determine rent location. A bottleneck produces durable rent only where replication barriers are rising, competitors cannot reproduce the capability, customers cannot substitute, and incumbents cannot absorb the value. That test is applied below and not restated.

Durable rent stays concentrated in EUV lithography, the leading-edge foundry, electronic design automation, and the leading accelerator ecosystem. Most energy and physical-infrastructure layers are strategically necessary but economically competitive. A middle group — high-bandwidth memory, grid equipment, the qualified power-device tier — earns the temporary returns of a capacity cycle. The battery and storage leader is the physical-layer exception.

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2. Evidence architecture for the core conclusions

Each top-level conclusion stated as claim, mechanism, evidence with confidence, and the condition that would invalidate it. Evidence is tagged observed (high), estimate (medium), or judgement (lower).

C1 — Constraint location is not rent location. Mechanism: a bottleneck attracts capital; capital relieves reproducible scarcity; rent survives only where the barrier rises as capital arrives. Evidence: power, transformers, and interconnection are the current binding constraints (estimate); generation, grid hardware, storage, and conversion are reproducible with capital and historically earn the cost of capital (observed, from utility and equipment economics). Confidence: high on the mechanism, medium on the current constraint location. Invalidation: a physical layer develops a proprietary standard or qualification barrier that capital cannot replicate.

C2 — Rent concentrates upstream in semiconductors. Mechanism: escalating capital and tacit yield-learning raise the leading-edge barrier faster than entrants can climb it, while a single lithography tool and a concentrated design-tool layer have no substitute. Evidence: lithography monopoly economics, ~60% foundry gross margin, ~80%+ design-tool gross margin (observed, FY2024–25). Confidence: high. Invalidation: AI value shifts downstream so integrators capture the margin and chokepoints become cost-plus.

C3 — Power and physical infrastructure commoditize. Mechanism: reproducible capacity competes returns toward cost even while demand is acute. Evidence: solar holds the majority of global capacity yet generates losses; a leading wide-bandgap substrate specialist entered insolvency in mid-2025 after a ~40% price collapse (observed); grid-equipment pricing power is real but tied to a capacity cycle (estimate). Confidence: high on the pattern, medium on duration. Invalidation: a proprietary power-delivery standard with lock-in forms.

C4 — China shows high capability and divergent returns. Mechanism: a system optimizing for resilience and capability does not optimize for return on capital. Evidence: dominant solar capacity with losses, a subsidized sub-economic leading edge, and a single profitable physical-layer participant in batteries (observed). Confidence: high on capability, high on the return divergence, lower on forward trajectory. Invalidation: domestic frontier ecosystems become self-sustaining and economically competitive.

C5 — Accelerator rent is durable but less certain than equipment monopolies. Mechanism: software lock-in and full-stack co-design raise switching costs, but architecture markets evolve faster than equipment markets. Evidence: ~75% gross margin and an intact installed base (observed); growing hyperscaler custom-silicon programs (estimate). Confidence: high on current economics, lower on duration. Invalidation: abstraction layers and captive silicon reduce dependence enough to migrate inference at scale.

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3. Rent model

Economic rent = Structural capability × Economic capture × Duration, normalized and scaled to 0–10. The score measures business quality and durability, not investment return.

Duration benchmark. **9–10:** near-monopoly, barriers increasing, little substitution. **7–8:** strong moat with manageable geographic, customer, or technology risk. **5–6:** strong position with meaningful substitution, cycle, or customer risk. **3–4:** temporary pricing power. **1–2:** commodity economics.

Table A — Structural Capability (1–10 per factor)

Company / layer	Concentration	Replication	Moat	Qualification	Score /40
EUV lithography	10	10	10	10	40
Leading-edge foundry	8	9	9	9	35
Design tools (EDA)	8	8	9	7	32
Accelerator + software	8	7	10	7	32
HBM memory	7	7	7	8	29
Advanced packaging	7	7	7	7	28
Battery/storage leader	7	6	7	8	28
Semiconductor equipment	6	7	7	7	27
Grid / qualified power semis	6	6	5	7	24
HVDC / optical interconnect	5	6	5	5	21
Commodity energy / devices	3	3	2	3	11

Table B — Economic Capture (observed, approximate, FY2024–25; high confidence)

Company / layer	Gross margin	ROIC	Evidence /10
Leading-edge foundry	~60%	~32%	10
Accelerator + software	~75%	~75%	10
EUV lithography	~52%	~24%	9
Design tools (EDA)	~80%+	High	9
HBM memory	High (up-cycle)	Cyclical	7
Advanced packaging	Healthy	High	7
Battery/storage leader	~24% (27% storage)	Solid	7

Company / layer	Gross margin	ROIC	Evidence /10
Semiconductor equipment	~45–48%	High	7
Grid / qualified power semis	~40%, improving	Moderate/cyclical	5–6
HVDC / optical interconnect	Pre-rent	Low	4
Commodity energy / devices	Negative–~2%	Neg–low	1–2

Table C — Final Classification (lower confidence on duration)

Company / layer	Duration (band, binding risk)	Final /10	Archetype
EUV lithography	9 — served-market concentration	**8.1**	Structural monopoly
Leading-edge foundry	7 — geographic concentration	**6.1**	Durable oligopoly
Design tools (EDA)	8 — oligopoly, not monopoly	**5.8**	Durable oligopoly
Accelerator + software	6 — internal ASIC / buyer power	**4.8**	Durable oligopoly (valuation caveat)
Battery/storage leader	6 — commoditization	**2.9**	Selective exception
Semiconductor equipment	6 — capex cyclicality	**2.8**	Durable oligopoly
Advanced packaging	5 — incumbent absorption	**2.5**	Bottleneck, incumbent-held
HBM memory	4 — cyclicality	**2.0**	Cyclical pricing power
Grid / qualified power semis	4 — super-cycle / scaling	**1.4**	Cyclical pricing power
HVDC / optical interconnect	3 — incumbent absorption	**0.6**	Ownership uncertain
Commodity energy / devices	2 — commodity economics	**0.1**	Scale without rent

4. Semiconductor rent analysis

ASML. Source of rent: lithography monopoly. Replication barrier: extreme, on a three-decade supplier ecosystem. Customer dependence: total, no alternative. Substitution risk: low; alternative lithography not at commercial scale. Absorption risk: none. Evidence quality: high. Duration 9. Falsification: alternative lithography reaches commercial scale.

TSMC. Source: demand-aggregation and yield-learning flywheel. Replication: high, knowledge tacit. Customer dependence: high. Substitution risk: low. Absorption risk: low. Evidence quality: high. Duration 7 — held down by geographic concentration. Falsification: customer-driven second-sourcing or geographic diversification fragments the demand that funds the flywheel.

NVIDIA. Source: developer lock-in, networking, full-stack co-design. Replication: silicon partial, software not yet. Customer dependence: high but concentrated. Substitution risk: moderate — ASICs and abstraction layers. Absorption risk: moderate — captive silicon at the largest buyers. Evidence quality: high on current economics, lower on duration. Duration 6. Falsification: abstraction plus captive silicon migrate inference at scale.

EDA. Source: mid-design workflow lock-in. Replication: high. Customer dependence: high. Substitution risk: low; no full-flow alternative. Absorption risk: low. Evidence quality: high. Duration 8. Falsification: an open or integrated tool flow reaches advanced-node parity.

HBM. Source: bandwidth scarcity. Replication: moderate; three credible suppliers. Customer dependence: high near-term. Substitution risk: low near-term. Absorption risk: low. Evidence quality: high on cyclicality. Duration 4. Falsification: structural capacity discipline holds margins through a downcycle.

Advanced packaging. Source: capacity scarcity, held by the leading foundry. Replication: high. Customer dependence: high. Substitution risk: low. Absorption risk: high — the foundry internalizes the value. Evidence quality: medium. Duration 5. Falsification: an independent packaging specialist establishes a durable standard.

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5. Power, grid, battery, and energy

The conclusion is in C3. The specifics: value flows to existing holders — the leading accelerator franchise extending into power delivery through a power-semiconductor partnership, established equipment suppliers in transformers and grid conversion, and utilities in interconnection. Grid equipment earns real but capacity-cycle pricing power (3–4 band). The qualified silicon-carbide and gallium-nitride device tier holds brief oligopoly rent that erodes as capacity scales. The battery and storage leader is the exception, at ~17% net and ~27% gross margin in stationary storage, because that segment is concentrated, differentiated, and qualification-gated. For optical interconnect, HVDC, and new power architectures, monitor because technological transitions can create unexpected standards, not because independent rent is the current base case.

Capital cycle: capital attraction is not value creation. The 2026 operator capital expenditure is weighted toward the most reproducible layers, where returns normalize toward cost, while the high-barrier chokepoints compound. Solar's losses despite dominant capacity are the working case. Durable returns accrue to those controlling standards, ecosystems, qualification barriers, scarce knowledge, or irreplaceable tools — not those receiving the most investment.

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6. China analysis: three lenses

China's industrial system combines market competition, industrial policy, and strategic-capability objectives; it is neither purely political nor purely market-driven.

Industrial capability is strong and broad: manufacturing scale, battery leadership, solar deployment, mature semiconductor production, supply-chain integration. **Economic return** is uneven and largely the inverse: scale leadership does not create pricing power, so solar generates losses, the leading edge is sub-economic, and energy infrastructure earns commodity returns. **Strategic objective** is consistent and high; Five-Year Plans evidence priorities — supply security, technological independence, national capability — not economic success.

A system optimized for resilience and capability produces different ROIC outcomes from a shareholder-return-optimized system, because resilience and capability are not the same objective as return on capital. Battery and storage is the exception, where concentration, technology, qualification, and scale convert capability into rent.

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7. Three types of winners

Economic winners (durable rent). ASML, the leading-edge foundry, design tools, the accelerator franchise, and the battery and storage leader as the physical-layer exception.

Industry winners (strategic importance). A different set: the US AI ecosystem; the Taiwan-based leading-edge foundry ecosystem; European semiconductor equipment;

Korean memory; Japanese materials and equipment; the Chinese manufacturing and battery ecosystem. Strategic importance does not require rent.

Shareholder-return winners. Determined by valuation, expectations, entry price, and cycle position. A superior business can produce poor investment returns if expectations are excessive — most relevant to the accelerator franchise, where sustained demand and elevated margins are already capitalized. A cyclical business can produce strong returns when purchased below normalized economics. These three lists are not combined.

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8. Global industrial map

The stack is globally distributed; no economy controls the entire chain. United States: AI architecture, software, and semiconductor design, physically constrained at home by offshore manufacturing concentration and grid limits. Taiwan-based leading-edge foundry ecosystem: the most concentrated manufacturing position, carrying single-point supply risk. Europe: the lithography monopoly and specialized industrial technology, the deepest single barrier. Korea and Japan: Korean memory including HBM; Japanese materials, equipment, and components. China: manufacturing scale, battery leadership, and deployment capability, frontier-constrained per Section 6. Rent concentrates in a few links; physical scale distributes widely.

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9. Rent migration map

Historical: electricity rent sat in equipment and standards while generation commoditized; internet rent sat in platforms while transport commoditized; semiconductor rent concentrated at the tool, foundry, and design layers while mature chips commoditized. Current: EUV, foundry, design tools, and the accelerator ecosystem, with HBM and grid equipment earning cyclical returns. Possible future migration, where evidence emerges: the AI software-abstraction layer, new computing architectures, and proprietary energy systems. Low-probability migration: generic power infrastructure, commodity generation, and undifferentiated hardware.

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10. Falsification and quarterly monitoring indicators

Each major thesis with the condition that breaks it, the indicators to track each quarter, and the resulting action.

Accelerator ecosystem rent. Failure condition: abstraction layers and hyperscaler ASICs reduce dependence on the incumbent stack enough to migrate inference at scale.

Monitoring indicators: hyperscaler custom-silicon share of internal AI compute and the capital allocated to it; the leading franchise's revenue concentration among its largest buyers; the share of new inference deployments that are hardware-agnostic versus locked to the incumbent stack; the franchise's data-center gross-margin trend; and the move from allocation to open availability and pricing normalization. Investment implication: sustained margin compression plus rising ASIC share plus falling customer dependence → reclassify from durable oligopoly toward contested; rotate rent exposure toward the abstraction layer and captive-silicon enablers.

Power infrastructure. Failure condition: a proprietary power-delivery or energy architecture with qualification barriers and lock-in consolidates. Monitoring indicators: whether a single high-voltage DC and rack-power reference architecture becomes mandated across the dominant accelerator platform; the formation of patents and qualification barriers in data-center power delivery and who holds the reference IP; the depth of the leading franchise's vertical integration into power; and transformer and turbine lead times and order backlogs as the cyclical-relief signal. Investment implication: standard consolidation plus qualification barriers plus pricing that sustains rather than normalizes → reclassify HVDC and power delivery from MONITOR toward OWN; otherwise continue to treat grid equipment as a trade and exit on lead-time normalization.

China frontier capability. Failure condition: domestic equipment, design tools, HBM, and yield-learning become self-sustaining and economically competitive. Monitoring indicators: domestic HBM output and whether it reaches frontier bandwidth (the current binding constraint on domestic advanced-accelerator volume); domestic leading-edge yield and whether its cost economics approach commercial viability rather than remaining subsidized; domestic advanced-lithography progress toward production; full-flow domestic design-tool adoption at advanced nodes; and the localization rate of advanced etch, deposition, and metrology equipment. Investment implication: domestic frontier HBM plus improving leading-edge yield economics plus full-flow domestic tools → the frontier-constraint thesis weakens and a second pole forms; reduce the assumed durability of leading-edge rent concentration.

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11. Scenario framework

Probabilities are judgement (lower confidence).

Base case — rent remains concentrated upstream (~55%). Triggering evidence: continued lithography, foundry, and HBM scarcity; the accelerator franchise's margins hold; power stays reproducible and incumbent-captured; China stays frontier-constrained. Winners: ASML, the leading-edge foundry, design tools, the accelerator franchise, HBM cyclically, and the battery leader. Losers: commodity energy and manufacturing, and independent power-infrastructure entrants.

Bull case — rent migrates into new layers (~25%). Triggering evidence: a power-delivery or interconnect standard consolidates with qualification barriers, or system integration captures value downstream from components. Winners: the holder of the new standard (an incumbent extending its position, or an independent if capture fails), power-delivery and interconnect leaders, and system integrators. Losers: pure component chokepoints if value shifts downstream, and grid commoditizers if power standardizes around one architecture.

Bear case — current chokepoints weaken (~20%). Triggering evidence: CUDA dependence erodes through ASIC migration; demand normalization relieves scarcity; alternative lithography reaches scale; a China frontier breakthrough forms a second pole; or AI monetization lags capital expenditure and the capital cycle compresses returns. Winners: hyperscaler captive silicon, abstraction-layer owners, compute buyers, and cyclical names bought at trough. Losers: the accelerator rent position, the upstream-concentration thesis, and anyone who capitalized cyclical scarcity as durable.

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12. Evidence quality and final classification

High confidence (observed): current market structure, the FY2024–25 financials in Table B, existing monopolies and oligopolies. Medium confidence (estimate): power-demand and infrastructure forecasts — the ~945 TWh by 2030 and the ~$700bn 2026 capital-expenditure figures. Lower confidence (judgement): duration scores, scenario probabilities, and forward rent-migration locations. The structural classifications are firmer than the duration estimates; the forward-transition calls are the least certain element.

Final classification — a business-quality classification, not a stock recommendation; security selection requires a valuation and expectations overlay.

OWN — durable rent. EUV lithography; the leading-edge foundry; design tools; the equipment oligopoly; the accelerator franchise, with lower duration certainty and the valuation caveat. The battery and storage leader is the selective exception.

TRADE — temporary pricing power. HBM, grid equipment, and the qualified power-device tier — to the capacity-catch-up curve.

MONITOR — potential future rent, currently unproven. Optical interconnect, silicon photonics, advanced packaging, HVDC, AI energy management — tracked against the Section 10 indicators.

AVOID — commodity economics. Most energy infrastructure and commodity manufacturing — solar, generic cells, commodity substrate, standalone cooling; and infrastructure credit priced as though collateral does not decay.

Figures approximate and current to Q2 2026: observed financials high confidence, industry estimates medium, duration, scenario, and forward-rent judgements lower. Geography treated as manufacturing concentration and supply-chain risk only. Forward statements are not forecasts or recommendations; independent verification required before use.